

K-th Smallest Number after Removal

You will be given N cards with integer x written on each card and Q queries to remove a card from your hand. After each time a card is removed, you need to output the k -th smallest number from the cards in your hand.

Input Format

- The first line consists of integer N , integer Q , and integer k .
- The second line consist of N unique numbers.
- The next Q lines consists of a integer x per line to remove.

Output Format

Output Q lines. For each query, output an integer that represents the k -th smallest number from the cards in your hand.

Constraints

- $1 \leq N, Q, k \leq 10000$.
- $N \geq Q + k$.
- Implement your algorithm in C using AVL tree concept for insertion and removal.

Sample Input 1 (standard input)

```
5 2 3
19 11 15 10 9
11
15
```

Sample Output 1 (standard output)

```
15
19
```

Sample Input 2 (standard input)

```
6 3 2
8 3 2 4 7 6
2
3
8
```

Sample Output 2 (standard output)

```
4
6
6
```

